$$Ab_2 = \begin{bmatrix}2 & 5\\ -3 & 1\end{bmatrix}\begin{bmatrix}-5\\ K\end{bmatrix} = \begin{bmatrix}2\cdot(-5)+5\cdot K\\ (-3)\cdot(-5)+1\cdot K\end{bmatrix} = \begin{bmatrix}5K-10\\ K+15\end{bmatrix}$$

(Row-Vector Rule)

$$\therefore AB = \begin{bmatrix}23 & 5K-10\\ -9 & K+15\end{bmatrix} \qquad BA = \begin{bmatrix}Ba_1 & Ba_2\end{bmatrix}$$ (By defn. of matrix multiplication)

$$Ba_1 = \begin{bmatrix}4 & -5\\ 3 & K\end{bmatrix}\begin{bmatrix}2\\ -3\end{bmatrix} = 2\begin{bmatrix}4\\ 3\end{bmatrix} + (-3)\begin{bmatrix}-5\\ K\end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix}8\\ 6\end{bmatrix} + \begin{bmatrix}15\\ -3K\end{bmatrix}$$ (By defn. of scalar multiple)

$$= \begin{bmatrix}23\\ 6-3K\end{bmatrix}$$ (Vector sum definition)

$$\cancel{Ba_2 = \begin{bmatrix}4 & -5\\ 3 & K\end{bmatrix}\begin{bmatrix}-5\\ K\end{bmatrix} = \begin{bmatrix}4\cdot(-5)+(-5)\cdot K\\ 3\cdot(-5)+K\cdot K\end{bmatrix}}$$

$$Ba_2 = \begin{bmatrix}4 & -5\\ 3 & K\end{bmatrix}\begin{bmatrix}5\\ 1\end{bmatrix} = 5\begin{bmatrix}4\\ 3\end{bmatrix} + 1\begin{bmatrix}-5\\ K\end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix}20\\ 15\end{bmatrix} + \begin{bmatrix}-5\\ K\end{bmatrix}$$ (By defn. of scalar multiple)

$$= \begin{bmatrix}15\\ 15+K\end{bmatrix}$$ (Vector sum definition)

$$\therefore BA = \begin{bmatrix}23 & 15\\ 6-3K & 15+K\end{bmatrix} \qquad AB = BA \Rightarrow \begin{bmatrix}23 & 5K-10\\ -9 & K+15\end{bmatrix} = \begin{bmatrix}23 & 15\\ 6-3K & 15+K\end{bmatrix}$$

$$5K-10 = 15 \Rightarrow K = 5 \qquad 6-3K = -9 \Rightarrow K = 5$$

10. Let $A = \begin{bmatrix}2 & -3\\ -4 & 6\end{bmatrix}$, $B = \begin{bmatrix}8 & 4\\ 5 & 5\end{bmatrix}$, and $C = \begin{bmatrix}5 & -2\\ 3 & 1\end{bmatrix}$.

Verify that $AB = AC$ and yet $B \neq C$.

$\boxed{\text{Solution:}}$ It's easy to see that $B \neq C$. $AB = \begin{bmatrix}2 & -3\\ -4 & 6\end{bmatrix}\begin{bmatrix}8 & 4\\ 5 & 5\end{bmatrix} = \begin{bmatrix}2\cdot 8+(-3)\cdot 5 & 2\cdot 4+(-3)\cdot 5\\ (-4)\cdot 8+6\cdot 5 & (-4)\cdot 4+6\cdot 5\end{bmatrix}$

(Row-Column Rule for Computing AB)

$$= \begin{bmatrix}1 & -7\\ -2 & 14\end{bmatrix}$$